



Performance optimization of sensorless control for PMSM in high-frequency pulse injection

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Abstract

For the sensorless control systems based on high-frequency (HF) signal injection in zero and low speed range, issues of harsh audible noise occur, as well as large estimation errors within the stages of acceleration and deceleration in ramp. To address the two problems, this paper proposes corresponding optimization methods. First, to remove the harsh discrete components in the responsive current and make the noise quiet and consistent, a single pair of complementary random HF pulses is injected only into the rotating estimated γ axis. Furthermore, an enhanced phase-locked loop (PLL) with a straightforward construction and simple parameter tweaking is proposed, which can eliminate transient errors completely. The accuracy and efficacy of the proposed approaches have been validated through simulation and experimental verification. Experimental results demonstrate that the proposed PLL can reduce the position error from 20° and 17.4° to zero under no-load and rated load, respectively. In addition, during the sudden removal of rated load at zero speed, the position error is reduced from 5.6° to 4.6° .

Keywords Enhanced phase-locked loop (EPLL) · Permanent magnet synchronous motor · Random pulse sequence high-frequency injection (HFPI-RS) · Sensorless control

1 Introduction

Permanent magnet synchronous motors (PMSMs) not only retain the core advantages of electrically excited synchronous motors [1, 2] but also possess characteristics such as high power density, high efficiency, simple structure, and easily implementable field-weakening operations, which are irreplaceable by other types of motors [3, 4]. The high-performance control of PMSM requires accurate rotor position information. However, the deficiencies of conventional encoder or resolver in signal reliability, environmental adaptability, installation, maintenance, cost, and debase affect the robustness, reliability, and product competitiveness of the drive system [5, 6]. For these reasons, sensorless

control technology was proposed and developed into a hotspot in the field of electric drive.

The methods to achieve the sensorless control are model-based method and signal injection method, which are suitable for medium-high and zero-low speed, respectively. When the speed is below 10% of the rated speed, the back electromotive force is relatively small, which causes difficulty in accurate estimation. Especially, the model-based system is unobservable at zero speed, and signal injection is generally adopted [7]. However, due to the injection of additional signals, the drive system will have problems of torque ripple, audible noise, system delay, and reduced control accuracy. The signal injection method is rooted in the saliency of motor. When the anisotropy ratio is lower than 1.2, closed-loop control is difficult to achieve using the signal injection. For solving these problems, [8, 9] used random signal injection to smooth the harsh noise of fixed frequency injection. In [10], standing in the point of suppressing torque ripple, a square wave signal with half sampling frequency is injected into the estimated q -axis to counteract the torque ripple generated by the high-frequency (HF) square wave signal injected into the d -axis for estimating the rotor position. The sensorless control of salient PMSM with a

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saliency ratio of approximately 1.1 is realized by current oversampling method in [11].

HF signals can be injected in the two-phase stationary reference frame ($\alpha\beta$ frame) and the estimated d axis (γ axis). The former is called rotating injection, and the latter is pulsating injection. Detailed comparison was conducted in [12, 13], which indicates that the pulsating injection has more advantages in system delay, estimation accuracy, torque ripple, audible noise, and adaptability to SPMSM. Similarly, HF current demodulation can also be performed in $\alpha\beta$ and $\gamma\delta$ frames. In general, the position estimation in the $\alpha\beta$ frame is based on the real position of the rotor. Thus, the closed-loop operation is simple, but the anti-interference is relatively poor. The $\gamma\delta$ -frame estimation is induced by the position error. As a result, it has high immunity to interference, and the closed loop is relatively troublesome. Thus, this study adopts pulsating injection and $\alpha\beta$ -frame current demodulation.

The signal forms of HF injection are sine wave, square wave, and pulse wave. In [14], pulse injection is employed, which avoids the use of any filters and position compensation. The pulse wave is equivalent to the combination of square wave and zero injection voltage. The introduction of zero voltage gives the time to HF responsive current to decay to zero. It also avoid the occurrence of DC offset, which improves the accuracy of signal demodulation and estimation. Compared with the first two, the pulse wave has the simplest and most flexible form, which is conducive to signal demodulation; this process reduces delay and provides better dynamic performance [8, 15]. Meanwhile, compared with the half sampling frequency square wave injection, the frequency of pulse injection is reduced, which will sacrifice a part of the current-loop bandwidth [15].

Phase-locked loop (PLL) is an important component of sensorless control, and its performance directly affects the quality of the estimated speed and position. Numerous studies have also centered on PLL, and Chen et al. optimized the PLL based on the finite control set [16] and Newton Raphson method [17]. However, a large error in the conventional quadrature PLL (QPLL) occurs during the stage of ramp acceleration and deceleration. To solve the problems of harsh noise in HF injection and QPLL transient errors, this study proposes the following two optimization measures.

1) A pair of complementary HF pulse sequences is randomly injected to eliminate discrete components in the current and smooth out harsh HF noise.

2) To address the limitation of QPLL, this study proposes a Type III enhanced PLL (EPLL). The proposed EPLL can reduce estimation errors to zero, offers easy parameter tuning, and requires only minor modifications to the QPLL structure.

2 Random pulse sequence high-frequency injection

A method called random pulse sequence high-frequency injection (HFPI-RS) is used to enhance PMSM sensorless control. Significant estimating errors and severe auditory noise during the ramp acceleration and deceleration stages are addressed by this approach.

2.1 HF model of PMSM

Similar to the fundamental frequency model, when HF voltage is injected into PMSM, it will induce HF current. When the frequency of voltage and current is much higher than the fundamental frequency, usually several kilohertz, the resistance voltage drops and items related to fundamental frequency can be ignored, especially when running at zero and low speed. In this case, the HF model can be simplified as

$$\begin{bmatrix} u_{dh} \\ u_{qh} \end{bmatrix} = \begin{bmatrix} L_{dh} & 0 \\ 0 & L_{qh} \end{bmatrix} \cdot \frac{d}{dt} \begin{bmatrix} i_{dh} \\ i_{qh} \end{bmatrix} \quad (1)$$

where u_{dh} , i_{dh} , L_{dh} , u_{qh} , i_{qh} , L_{qh} are d -, q -axis high frequency voltage, current, and inductance, respectively.

Based on the comparison in the Introduction, this study only injects HF voltage on the γ axis and demodulate HF current in the $\alpha\beta$ frame. The following appropriate transformations on (1) need to be conducted:

$$\begin{bmatrix} \cos \hat{\theta}_e & -\sin \hat{\theta}_e \\ \sin \hat{\theta}_e & \cos \hat{\theta}_e \end{bmatrix} \cdot \begin{bmatrix} 1/L_{dh} & 0 \\ 0 & 1/L_{qh} \end{bmatrix} \cdot \begin{bmatrix} \cos \tilde{\theta}_e & \sin \tilde{\theta}_e \\ -\sin \tilde{\theta}_e & \cos \tilde{\theta}_e \end{bmatrix} \begin{bmatrix} u_{\gamma h} \\ u_{\delta h} \end{bmatrix} = \frac{d}{dt} \begin{bmatrix} i_{\gamma h} \\ i_{\delta h} \end{bmatrix} \quad (2)$$

$$\frac{d}{dt} \begin{bmatrix} i_{\alpha h} \\ i_{\beta h} \end{bmatrix} = \begin{bmatrix} \cos \hat{\theta}_e & -\sin \hat{\theta}_e \\ \sin \hat{\theta}_e & \cos \hat{\theta}_e \end{bmatrix} \cdot \frac{d}{dt} \begin{bmatrix} i_{\gamma h} \\ i_{\delta h} \end{bmatrix} \quad (3)$$

(2) and (3) need to be simplified to obtain

$$\begin{bmatrix} \Delta i_{\gamma h} \\ \Delta i_{\delta h} \end{bmatrix} = \frac{t_s u_{\gamma h}}{L_{dh} L_{qh}} \begin{bmatrix} L_{qh} \cos^2 \tilde{\theta}_e + L_{dh} \sin^2 \tilde{\theta}_e \\ (L_{qh} - L_{dh}) \sin \tilde{\theta}_e \cos \tilde{\theta}_e \end{bmatrix} \quad (4)$$

$$\begin{bmatrix} \Delta i_{\alpha h} \\ \Delta i_{\beta h} \end{bmatrix} = \frac{t_s u_{\gamma h}}{L_{dh} L_{qh}} \sqrt{L_{qh}^2 \cos^2 \tilde{\theta}_e + L_{dh}^2 \sin^2 \tilde{\theta}_e} \cdot \begin{bmatrix} \cos(\theta_e - \varphi) \\ \sin(\theta_e - \varphi) \end{bmatrix} \quad (5)$$

$$\varphi = \arctan(L_{dh}/L_{qh} \tan \tilde{\theta}_e) \approx L_{dh}/L_{qh} \tilde{\theta}_e, (\text{when } \tilde{\theta}_e \approx 0) \quad (6)$$

where $i_{\gamma h}$, $i_{\delta h}$, $u_{\gamma h}$, $u_{\delta h}$, $i_{\alpha h}$, and $i_{\beta h}$ are the HF currents and voltages in the $\gamma\delta$ and $\alpha\beta$ frames, respectively. t_s is the sampling frequency. $\Delta i_{\gamma h}$, $\Delta i_{\delta h}$, $\Delta i_{\alpha h}$, and $\Delta i_{\beta h}$ are the corresponding HF current changes within the unit sampling interval. $\tilde{\theta}_e = \theta_e - \hat{\theta}_e$ is position error, and θ_e , $\hat{\theta}_e$ represents the actual and estimated positions of the rotor.

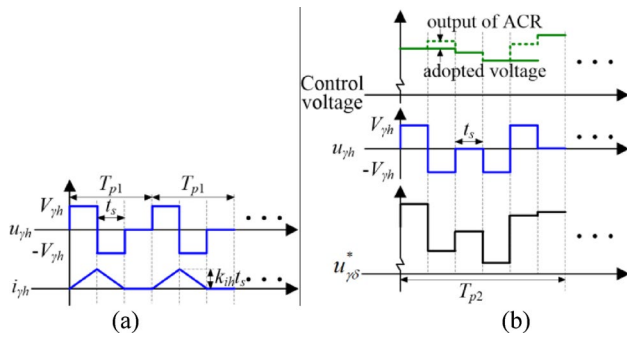


Fig. 1 Signal injection and processing: **a** HF voltage and responsive current of HFPI-FS; **b** selection of control voltage and synthesis of reference voltage

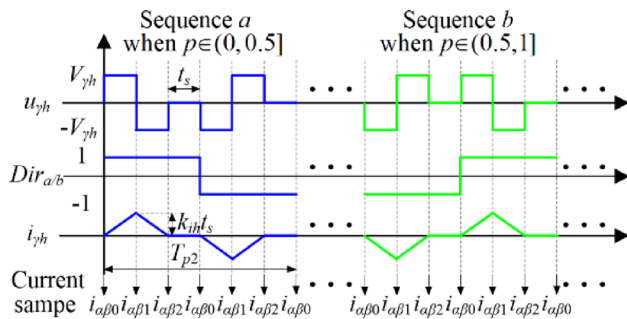


Fig. 2 HF injection voltage and responsive current of HFPI-RS

In (5), position information is included, and estimated speed and position can be obtained through PLL.

2.2 HFPI-FS and PSD

For fixed sequence HF pulse injection (HFPI-FS), the HF voltage is shown in Fig. 1a. When the system tends to stabilize and $\theta_{err} \approx 0$, the change in HF responsive current is

$$\begin{bmatrix} \Delta i_{\gamma h} \\ \Delta i_{\delta h} \end{bmatrix} \approx \frac{t_s u_{\gamma h}}{L_{dh}} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad (7)$$

$i_{\gamma h}$ can be approximated linearly as (8), as shown in Fig. 1a.

$$i_{\gamma h_FS} = \begin{cases} k_{ih} \cdot t_r(t, T_{p1}), & t_r(t, T_{p1}) \in (0, T_{p1}/3] \\ -k_{ih} \cdot t_r(t, T_{p1}) + 2k_{ih}T_{p1}/3, & t_r(t, T_{p1}) \in (T_{p1}/3, 2T_{p1}/3] \\ 0, & t_r(t, T_{p1}) \in (2T_{p1}/3, T_{p1}] \end{cases} \quad (8)$$

where k_{ih} is $V_{\gamma h}/L_{dh}$, which is the rate of change of $i_{\gamma h}$. T_{p1} is the period of the pulse sequence, which is the triple sampling frequency, and $t_r(t, T_{p1})$ is the remainder of t divided by T_{p1} .

The PSD for the signal in the form of (8) is given as follows [18]:

$$S(f) = \begin{cases} \frac{1}{E[T_k]} \left\{ E[|I(f)|^2] + 2 \operatorname{Re} \left[\frac{E[I(f)]e^{j2\pi f T_k} \cdot E[I^*(f)]}{1 - E[e^{j2\pi f T_k}]} \right] \right\}, & E[e^{j2\pi f T_k}] \neq 1 \\ \frac{|E[I(f)]|^2}{\{E[T_k]\}^2}, & E[e^{j2\pi f T_k}] = 1 \end{cases} \quad (9)$$

where $E[\cdot]$ is the mathematical expectation operator, and $I(f)$ is the Fourier transform of the HF responsive current in the unit period. T_k is the period of the random periodic signal. For HFPI-FS, T_k is a fixed T_{p1} . As shown in (8), the PSD of HF current includes two parts: discrete and continuous. These terms correspond to whether $E[e^{j2\pi f T_k}]$ is equal to 1 or not.

The Fourier transform of (8) is

$$I_{FS}(f) = \int_0^{T_{p1}/3} k_{ih}t \cdot e^{-j2\pi f t} dt + \int_{T_{p1}/3}^{2T_{p1}/3} (-k_{ih}t + 2k_{ih}T_{p1}/3) e^{-j2\pi f t} dt \\ = -\frac{k_{ih}}{4\pi^2 f^2} (e^{-j2\pi f T_{p1}/3} - 1)^2 \quad (10)$$

(10) is substituted into (9) to acquire

$$S_{FS}(f) = \begin{cases} 0, & \text{continuous term} \\ 0, & \text{mod}(n, 3) = 0 \\ |I(f)|^2/T_{p1}^2, & \text{mod}(n, 3) \neq 0 \end{cases} \quad (11)$$

where n is a nonnegative integer, and $\text{mod}(n, 3)$ is the remainder of n divided by 3.

As shown in (11), the harmonics of HFPI-FS are mainly concentrated in non-third harmonic frequencies of the injection frequency, which will result in harsher audible noise.

2.3 Proposed HFPI-RS and PSD

According to (9), to avoid the harsh HF noise, the discrete component must be 0, that is,

$$E[I(f)] = 0 \quad (12)$$

If the probability of HF responsive current is equal, then the amplitude is the same and the direction is opposite, which satisfies (12). In practice, the probability of responsive current with the same amplitude and opposite direction is not always equal. As a result, the γ -axis HF current has a DC bias, which affects the demodulation of the position error information and reduces the estimation accuracy. Therefore, this study injects a complementary HF pulse voltage per unit period, as shown in Fig. 2.

For the random HF pulse voltage in Fig. 2, when the pseudo-random number is within the range of (0,0.5], pulse sequence *a* is used. When the pseudo-random number is within the range of (0.5,1], sequence *b* is used. The pseudo-random number is generated by a congruence generator.

Similar to HFPI-FS, for sequence a , its HF responsive current is

$$i_{\gamma ha_RS} = \begin{cases} k_{ih} \cdot t_r(t, T_{p2}), & t_r(t, T_{p2}) \in (0, T_{p2}/6] \\ -k_{ih} \cdot t_r(t, T_{p2}) + k_{ih}T_{p2}/3, & t_r(t, T_{p2}) \in (T_{p2}/6, T_{p2}/3] \\ -k_{ih} \cdot t_r(t, T_{p2}) + k_{ih}T_{p2}/2, & t_r(t, T_{p2}) \in (T_{p2}/2, 2T_{p2}/3] \\ k_{ih} \cdot t_r(t, T_{p2}) - 5k_{ih}T_{p2}/6, & t_r(t, T_{p2}) \in (2T_{p2}/3, 5T_{p2}/6] \\ 0, & \text{elsewhere} \end{cases} \quad (13)$$

where T_{p2} is the injection period, which is 6 times the sampling frequency.

Fourier transform is performed on (13), and its derivation process is similar to (10).

$$I_{aRS}(f) = \int_0^{T_{p2}} i_{\gamma ha_RS} \cdot e^{-j2\pi ft} dt \\ = -\frac{k_{ih}}{4\pi^2 f^2} e^{-j5\pi f T_{p2}/3} (e^{j\pi f T_{p2}/3} - 1)^3 (e^{j2\pi f T_{p2}/3} + e^{j\pi f T_{p2}/3} + 1) \quad (14)$$

Owing to

$$i_{\gamma hb_RS} = -i_{\gamma ha_RS} \quad (15)$$

and

$$I_{bRS}(f) = -I_{aRS}(f) \quad (16)$$

According to (14) and (16), the mathematical expectation of HF current can be obtained, which is

$$E[I_{RS}(f)] = 0.5I_{aRS}(f) + 0.5I_{bRS}(f) = 0 \quad (17)$$

Substituting (14), (16), and (17) into (9) yields

$$S_{RS}(f) = \begin{cases} |I_{aRS}(f)|^2/T_{p2}, & f \neq n/T_{p2} \\ 0, & f = n/T_{p2} \end{cases} \quad (18)$$

As shown in (18), HFPI-RS does not have a discrete component and does not produce harsh HF noise. However, the continuous component still exists, and the noise is stable, which is equivalent to smoothing the HF noise.

2.4 Signal demodulation and position estimation

In HF signal injection, owing to the existence of FOC command variations, inverter nonlinearities, resistance, and rotating voltage drops, the actual injection voltage will be deviated, which can be written as

$$u_{\gamma} = \begin{bmatrix} u_{\gamma h} + V_{\gamma e} \\ V_{\delta e} \end{bmatrix} \quad (19)$$

where $V_{\gamma e}$ and $V_{\delta e}$ are the voltage errors of γ and δ axes. $u_{\gamma h} = \pm V_{\gamma h}$, and $V_{\gamma h}$ is the amplitude of the injected HF voltage.

Introducing (19) into (5), and similarly considering the coordinate transformation corresponding to $u_{\delta h}$, the current variation in unit period can be obtained as (20).

In (20), the control current remains constant within a unit sampling time. According to (20), conventional HF current demodulation based on current variation will contain voltage errors, which reduces the accuracy of position estimation. Therefore, this study adopts the given FOC voltage holding strategy proposed in [19]. Within the first non-zero and zero injection pulse voltage, the given voltage is updated by the current regulator. In the second non-zero injection pulse, the given voltage keeps the same as the given voltage during the first non-zero injection pulse, as shown in Fig. 1b. The HF voltage error, true position, and position error are assumed to remain unchanged during two adjacent injection times. As a result, we can obtain (21).

$$\begin{cases} \Delta i_{\alpha 10} = \Delta i_{\alpha 10} = [(L_{qh} \cos \theta_e \cos \tilde{\theta}_e + L_{dh} \sin \theta_e \sin \tilde{\theta}_e)(V_{\gamma h} + V_{\gamma e}) \\ \quad + (L_{qh} \cos \theta_e \sin \tilde{\theta}_e - L_{dh} \sin \theta_e \cos \tilde{\theta}_e)V_{\delta e}]t_s/L_{dh}L_{qh} \\ \Delta i_{\beta 10} = \Delta i_{\beta 10} = [(L_{qh} \sin \theta_e \cos \tilde{\theta}_e - L_{dh} \cos \theta_e \sin \tilde{\theta}_e)(V_{\gamma h} + V_{\gamma e}) \\ \quad + (L_{qh} \sin \theta_e \sin \tilde{\theta}_e + L_{dh} \cos \theta_e \cos \tilde{\theta}_e)V_{\delta e}]t_s/L_{dh}L_{qh} \end{cases} \quad (20)$$

$$\begin{cases} \Delta i_{\alpha 21} = [(L_{qh} \cos \theta_e \cos \tilde{\theta}_e + L_{dh} \sin \theta_e \sin \tilde{\theta}_e)(-V_{\gamma h} + V_{\gamma e}) \\ \quad + (L_{qh} \cos \theta_e \sin \tilde{\theta}_e - L_{dh} \sin \theta_e \cos \tilde{\theta}_e)V_{\delta e}]t_s/L_{dh}L_{qh} \\ \Delta i_{\beta 21} = [(L_{qh} \sin \theta_e \cos \tilde{\theta}_e - L_{dh} \cos \theta_e \sin \tilde{\theta}_e)(-V_{\gamma h} + V_{\gamma e}) \\ \quad + (L_{qh} \sin \theta_e \sin \tilde{\theta}_e + L_{dh} \cos \theta_e \cos \tilde{\theta}_e)V_{\delta e}]t_s/L_{dh}L_{qh} \end{cases} \quad (21)$$

(21) is subtracted from (20) to obtain

$$\begin{bmatrix} \Delta i_{\alpha 10} - \Delta i_{\alpha 21} \\ \Delta i_{\beta 10} - \Delta i_{\beta 21} \end{bmatrix} = \frac{2t_s V_{\gamma h}}{L_{dh}L_{qh}} \sqrt{L_{qh}^2 \cos^2 \tilde{\theta}_e + L_{dh}^2 \sin^2 \tilde{\theta}_e} \begin{bmatrix} \cos(\theta_e - \varphi) \\ \sin(\theta_e - \varphi) \end{bmatrix} \quad (22)$$

Similarly, in the latter half of the injection cycle, the direction of the difference in current change is negative. For injection sequence a , the current difference can be written uniformly as.

$$Dir_a \cdot \begin{bmatrix} \Delta i_{\alpha 10} - \Delta i_{\alpha 21} \\ \Delta i_{\beta 10} - \Delta i_{\beta 21} \end{bmatrix} = \frac{2t_s V_{\gamma h}}{L_{dh}L_{qh}} \sqrt{L_{qh}^2 \cos^2 \tilde{\theta}_e + L_{dh}^2 \sin^2 \tilde{\theta}_e} \begin{bmatrix} \cos(\theta_e - \varphi) \\ \sin(\theta_e - \varphi) \end{bmatrix} \quad (23)$$

where Dir_a is the directional function of the current difference, which is defined as follows:

$$Dir_a = \begin{cases} 1, & t_r(t, T_{p2}) \in (0, T_{p2}/2] \\ -1, & t_r(t, T_{p2}) \in (T_{p2}/2, T_{p2}] \end{cases} \quad (24)$$

For injection sequence b , the direction function is opposite to sequence a , which is

$$Dir_b(t, T_{p2}) = -Dir_a(t, T_{p2}) \quad (25)$$

The position information obtained by (23) contains some noise, which is filtered out by an adaptive low-pass filter (ALPF) in this study. When the estimated frequency is below 7.5 Hz, the cutoff frequency is fixed at 150 Hz. When the estimated frequency is higher than 7.5 Hz, the cutoff frequency is set to 20 times the estimated frequency, and the mathematical expression is given as follows:

$$\begin{bmatrix} \Delta i_{\cos} \\ \Delta i_{\sin} \end{bmatrix} = \text{ALPF} \begin{bmatrix} \text{Dir}_{a/b} \cdot (\Delta i_{\alpha 10} - \Delta i_{\alpha 21}) \\ \text{Dir}_{a/b} \cdot (\Delta i_{\beta 10} - \Delta i_{\beta 21}) \end{bmatrix} \quad (26)$$

To avoid the influence of coefficient changes in (23) on the parameters of PLL, (26) is normalized.

$$\begin{bmatrix} \Delta i_{\cos} \\ \Delta i_{\sin} \end{bmatrix} = \frac{1}{\sqrt{(\Delta i_{\cos})^2 + (\Delta i_{\sin})^2}} \begin{bmatrix} \Delta i_{\cos} \\ \Delta i_{\sin} \end{bmatrix} = \begin{bmatrix} \cos(\theta_e - \varphi) \\ \sin(\theta_e - \varphi) \end{bmatrix} \quad (27)$$

3 Enhanced phase-locked loop

PLL is a crucial component in the sensorless control of PMSMs. Its performance directly impacts the quality of the estimated speed and position. This section will discuss the conventional QPLL and proposed EPLL.

3.1 Conventional QPLL

Figure 3 shows the control block diagram of QPLL based on rotor position information, and the output of the phase detector is [20]

$$\begin{aligned} \Delta \theta_e &= \frac{L_{qh}}{L_{qh} - L_{dh}} (-\Delta i_{\cos n} \cdot \sin \hat{\theta}_e + \Delta i_{\sin n} \cdot \cos \hat{\theta}_e) \\ &= \frac{L_{qh}}{L_{qh} - L_{dh}} \sin[(\theta_e - \varphi) - \hat{\theta}_e] \approx \tilde{\theta}_e \end{aligned} \quad (28)$$

In Fig. 3, to ensure a unified analysis, $L_{qh}/(L_{qh}-L_{dh})$ is incorporated into the parameters of the loop filter (LF), which means

$$k_p = k'_p(L_{qh} - L_{dh})/L_{qh}, k_i = k'_i(L_{qh} - L_{dh})/L_{qh} \quad (29)$$

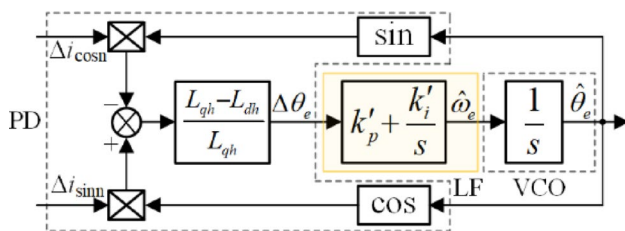


Fig. 3 Control block diagram of QPLL

The transfer function of the rotor position estimation error to the real position can be obtained, that is,

$$\frac{\tilde{\theta}_e}{\theta_e} = \frac{\tilde{\theta}_e}{\tilde{\theta}_e + \hat{\theta}_e} = \frac{1}{1 + \frac{\tilde{\theta}_e}{\hat{\theta}_e}} \approx \frac{1}{1 + \frac{k_p s + k_i}{s^2}} \approx \frac{s^2}{s^2 + k_p s + k_i} \quad (30)$$

In (30), when the position error is very small, $\Delta \theta_e \approx \tilde{\theta}_e$.

When the given speed is a ramp, the real rotor position θ_e is

$$\theta_e = \frac{a_c}{s^3} \quad (31)$$

where a_c is the slope of the given electrical angular velocity (in rad/s^2).

Combining (30) and (31), according to the final value theorem, the position estimation error, with the system tending to stabilize, can be obtained, that is,

$$\tilde{\theta}_e|_{t \rightarrow +\infty} = \lim_{s \rightarrow 0} \frac{s^2}{s^2 + k_p s + k_i} \cdot \frac{a_c}{s^3} \cdot s = \frac{a_c}{k_i} \quad (32)$$

As shown in (32), when the given speed is a ramp, an error always occurs in the estimated position. To reduce the error, the k_i value of the QPLL needs to be increased or the slope of the given speed should be reduced. For general applications, the acceleration a_c is relatively fixed. In (31), $\tilde{\theta}_e$ is inversely proportional to k_i . When k_i increases to a certain value, the decrease of $\tilde{\theta}_e$ is very small, and too large k_i will produce very heavy noise and vibration.

3.2 Proposed EPLL

According to (32), to guarantee that $\tilde{\theta}_e$ is zero, the numerator of the PLL transfer function must be s^3 . If so, then the transfer function of PLL can be built as

$$\frac{\tilde{\theta}_e}{\theta_e} = \frac{s^3}{s^3 + As^2 + Bs + C} \quad (33)$$

where A , B , and C are constants to be determined.

According to (33), the open-loop transfer function from $\tilde{\theta}_e$ to $\hat{\theta}_e$ can be calculated backward. It is

$$\frac{\hat{\theta}_e}{\tilde{\theta}_e} = \frac{\theta_e - \tilde{\theta}_e}{\tilde{\theta}_e} = \frac{As^2 + Bs + C}{s^3} \approx \frac{\hat{\theta}_e}{\Delta \theta_e} \quad (34)$$

(34) is a Type III system (called EPLL). To provide higher stability margins [21], (34) is set to

$$\frac{\hat{\theta}_e}{\Delta \theta_e} = K \frac{(s + \omega_z)^2}{s^3} \quad (35)$$

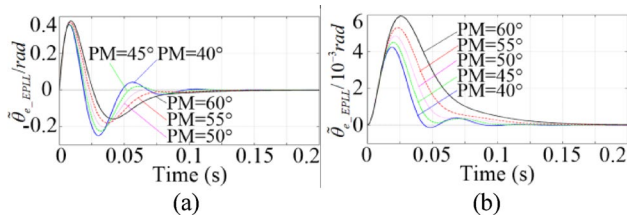


Fig. 4 Transient response of the rotor position error with various PM: **a** speed step input; **b** speed ramp input

where K is the open-loop gain of the system and $-\omega_z$ is its zero point.

For the open-loop transfer function of (34), the expression for its phase margin (PM) is

$$PM = 2 \arctan \frac{\omega_c}{\omega_z} - \frac{\pi}{2} \quad (36)$$

where ω_c is the cutoff frequency with the unit of rad/s , which satisfies

$$\omega_c = \frac{K}{\sin^2(2 \arctan \frac{\omega_c}{\omega_z})} \quad (37)$$

Combining (36) and (37), K , ω_z can be expressed as PM and ω_c with

$$\omega_z = \frac{\omega_c}{\tan(PM) + \sec(PM)} \quad (38)$$

$$K = \omega_c \frac{\sin(PM) + 1}{2} \quad (39)$$

From (38) and (39), the parameter tuning of EPLL can be realized as long as PM and ω_c are determined. In engineering control, to ensure the stability of the system, the PM is generally set within the range of 40° – 60° .

Figure 4 shows the transient response curve of the rotor position estimated error corresponding to different PM at a fixed cut-off frequency ($\omega_c = 175 \text{ rad/s}$). This curve is plotted by the inverse Laplace transform of (34), where the constant values of the step function and the ramp function are taken as 100. When PM is smaller, the dynamic performance of tracking rotor position is better, but the oscillation time is longer. When PM is larger, the dynamic performance reduces, which suggests over-damping characteristics and long convergence time. Figure 6a shows the amplitude frequency characteristic of EPLL under different PM values. A lower PM corresponds to a greater amplification gain of the closed-loop transfer function of (34), as well as a more intense transient response oscillation of EPLL. The same conclusion can be obtained as Fig. 4. To acquire satisfactory

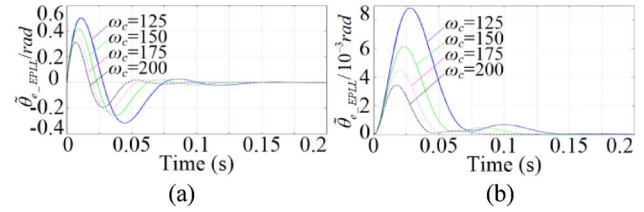


Fig. 5 Transient response of the rotor position error with various ω_c : **a** speed step input; **b** speed ramp input

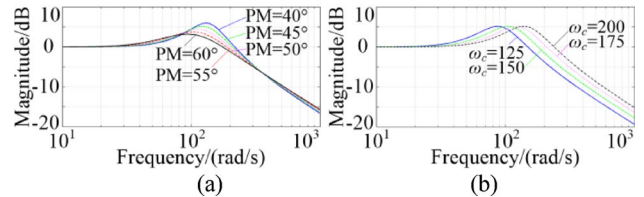


Fig. 6 Magnitude-frequency characteristic of EPLL: **a** different PM ($\omega_c = 175 \text{ rad/s}$); **b** different ω_c (PM = 45°)

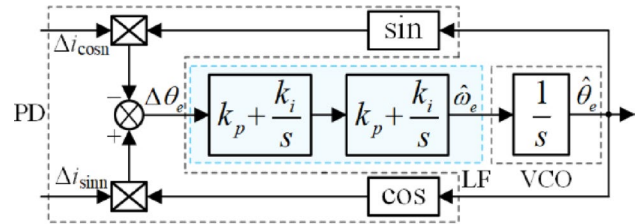


Fig. 7 Control block diagram of EPLL

dynamic performance, this study recommends the selection range of PM is 45° – 50° .

Figure 5 shows the transient response curve of rotor position error corresponding to different ω_c with fixed PM (PM = 45°). When ω_c is smaller, the dynamic performance of tracking rotor position is weaker. When ω_c is larger, the dynamic performance is improved and the convergence time is reduced. Figure 6b shows the Bode diagram of EPLL under different ω_c . When ω_c is smaller, the system has stronger anti-noise capability, but the dynamic performance is slower. With the increase in ω_c , the dynamic performance is improved, but the noise suppression capability to the middle frequency band is reduced. Therefore, the selection of ω_c should be based on the trade-off between dynamic performance and noise suppression capability.

In addition, for (35), the transfer function from $\Delta \theta_e$ to the estimated speed can be obtained by deformation, which is

$$\frac{\hat{\omega}_e}{\Delta \theta_e} = K \frac{(s + \omega_z)^2}{s^2} = (\sqrt{K} + \frac{\omega_z \sqrt{K}}{s})^2 \quad (40)$$

From (40), the LF of the EPLL is equivalent to two PIs with the same parameters in series, as shown in Fig. 7. Compared with the LF of QPLL, one PI is added.

4 Experimental results

According to the discussions in the previous two sections, a diagram of the sensorless control system based on HFPI-RS and EPLL in the $\gamma\delta$ frame can be obtained, as presented in Fig. 8. For simplicity, the $i_d = 0$ control framework is adopted. As mentioned earlier, pulses are stochastically injected into the estimated γ -axis. In the figure, p_n is the number of pole pairs. LPF and ALPF stand for low-pass filter and adaptive low-pass filter, respectively. To avoid the increased time delay caused by a filter with a fixed cutoff frequency, an ALPF is employed. The speed estimated by EPLL is used for the speed regulator, while the estimated position is utilized for coordinate transformation.

Based on the diagram and the actual hardware platform, a Simulink model was built, and the main parameters of the motor and hardware system are listed in Table 1. Given the limited space and the fact that the simulation waveforms are consistent with the experiments, no additional simulation results are provided here.

Figure 9 shows the experimental PMSM towing platform, which uses the TMS320F28335 as the main control chip. Notably, QDECCTL.SWAP is enabled and QPOCNT takes counting down as the positive direction in this platform. Therefore, the positive direction is the clockwise direction facing the motor shaft, as displayed in Fig. 9. The position in the experiment is the value that 360° minus the position in the previous analysis. The positions obtained by the encoder in the platform are only used to compare the estimation performance and not for closed-loop control. Moreover, for all experimental waveforms directly obtained via an oscilloscope, their zero scale or initial position is indicated by a right arrow marker with a number on the left side of the grid on the oscilloscope screen. This right arrow highlights the zero position by aligning with a certain horizontal grid line.

Table 2 presents the control-related parameters. ALL experimental parameters were fine-tuned based on simulation. To ensure a fair comparison between the QPLL and EPLL, which differ in topological structure, transfer functions, and number of parameters, both are configured with a comparable open-loop gain of 150. The cutoff frequency of the ALPF is set to 10 times the fundamental frequency. For frequencies below 7.5 Hz, a uniform cutoff frequency of 85 Hz is employed.

Figure 10 illustrates the experimental result of the a-phase current and normalized HF current difference under 300 rpm and rated load for the position system and the HFPI-RS sensorless system. Figure 11 shows the PSD and FFT of the corresponding a-phase current. In Fig. 10c, $\Delta i_{\cos n}$ and $\Delta i_{\sin n}$ have better sinuosity. As shown in

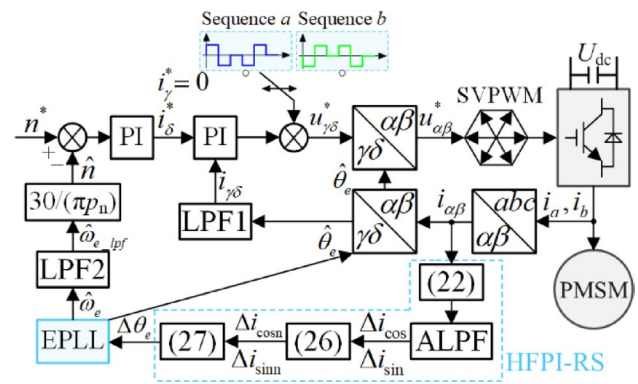


Fig. 8 Diagram of PMSM sensorless drive based on HFPI-RS and EPLL

Table 1 Main parameters of the PMSM experimental platform

Parameter	Value	Parameter	Value
Rated power	750 W	Pole pairs	5
DC-link voltage	311 V	Inductance (L_d/L_q)	6.04/9.06 mH
Rated torque	2.4 N·m	Stator resistance	1.45 Ω
Rated speed	3,000 r/min	PM flux linkage	0.12 Wb

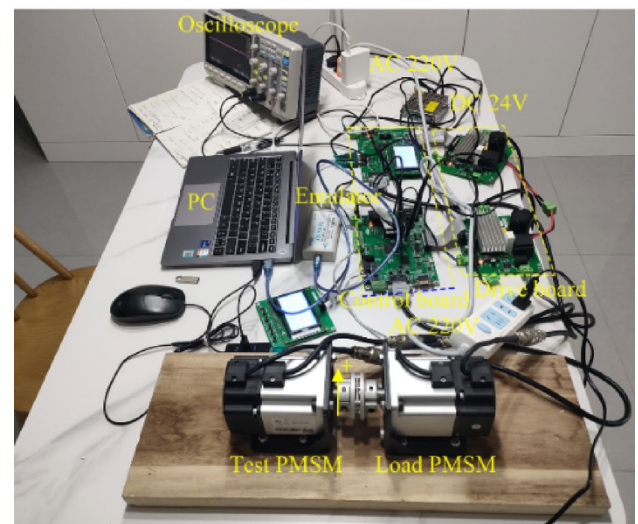


Fig. 9 PMSM towing platform

Table 2 Parameters related to the sensorless control algorithm

Parameter	Value	Parameter	Value
Phase Margin	45°	Cutoff frequencies of LPF1	400 Hz
Cutoff Frequency (ω_c)	175 rad/s	Cutoff frequencies of LPF2	350 Hz
k_p, k_i for EPLL in (40)	12.2/885	Adaptive coefficient of ALPF	10
k_p, k_i for QPLL in (29)	150/5625	Acceleration (a_c)	900 rpm/s
HF injection voltage	25 V	Carrier/Sampling frequency	10 kHz

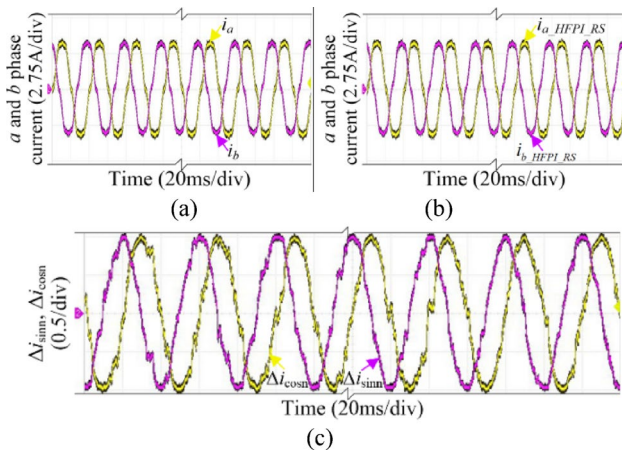


Fig. 10 *a*-phase currents and Δi_{coss} and Δi_{sinn} : **a** *a*-phase current of position system; **b** *a*-phase current of HFPI-RS sensorless system; **c** Δi_{coss} and Δi_{sinn}

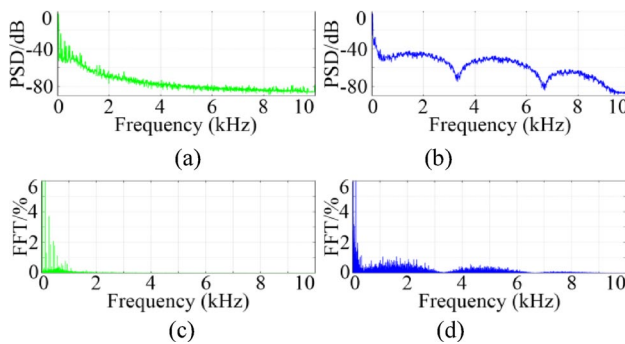


Fig. 11 PSD and FFT of *a*-phase currents: **a** PSD of position system; **b** PSD of HFPI-RS system; **c** FFT of position system; **d** FFT of HFPI-RS system

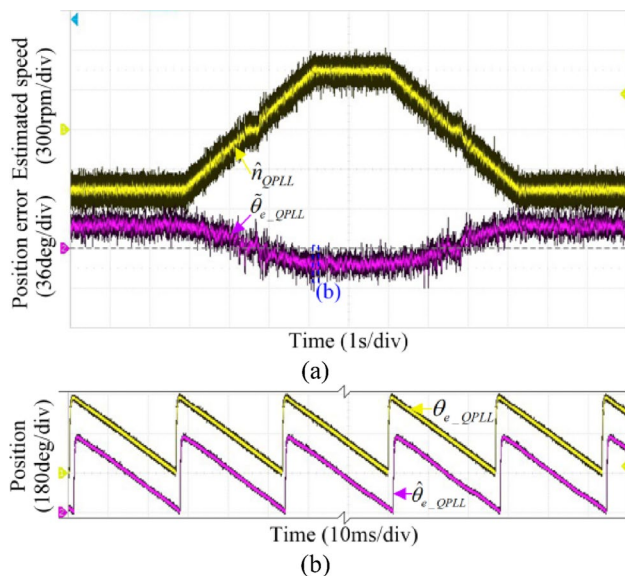


Fig. 12 Experimental results of QPLL system under no load and switching speed between -450 and 450 rpm: **a** estimated speed and position error; **b** actual and estimated position

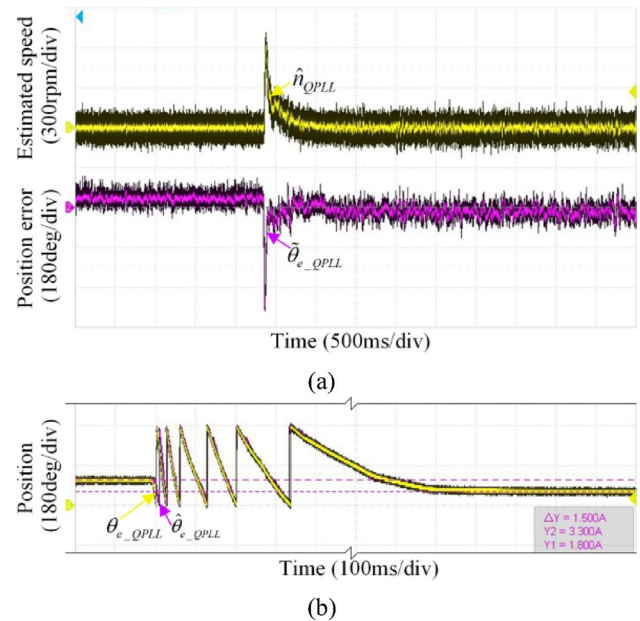


Fig. 13 Experimental result of QPLL system with suddenly unloading rated load at zero speed: **a** estimated speed and position error; **b** actual and estimated position

Fig. 11, HFPI-RS can effectively eliminate the discrete term in the responsive current, but larger harmonics still exist. The THD corresponding to Fig. 11d is 15.5%, while that to Fig. 11c is 10.8%. Owing to the consistent publication on HPFI-FS analysis and experimental results in [8, 22, 23], we only conducted simulation verification.

Figure 12 shows the estimated speed and position error of the sensorless system based on QPLL with switching speed between -450 and 450 rpm and no load. Transient errors occur during the stages of ramp acceleration and deceleration, which still exist after reaching steady state. The position errors are -14.8° at 450 rpm and 20° at -450 rpm.

Figure 13 demonstrates the estimated speed and position error of the QPLL sensorless system under the case of suddenly unloading rated load at zero speed. The position errors are 7.2° and 5.6° before and after unloading the load, respectively. Moreover, the position change where the rotor moves is 54° at zero speed.

Figure 14 shows the estimated speed, position error, and estimated position of the sensorless system based on QPLL under rated load and switching speeds between 0 and 450 rpm. Similar to the no-load situation, an estimation error occurs within the transient stage of acceleration and deceleration in ramp, which persists until the steady state. The position errors are 8.6° at 0 rpm, -15.8° at 450 rpm, and 17.4° at -450 rpm.

For comparing with the QPLL sensorless system, Figs. 15, 16 and 17 illustrate the experimental results of the same estimated variable under the same operating conditions. The function of EPLL is very obvious. Whether

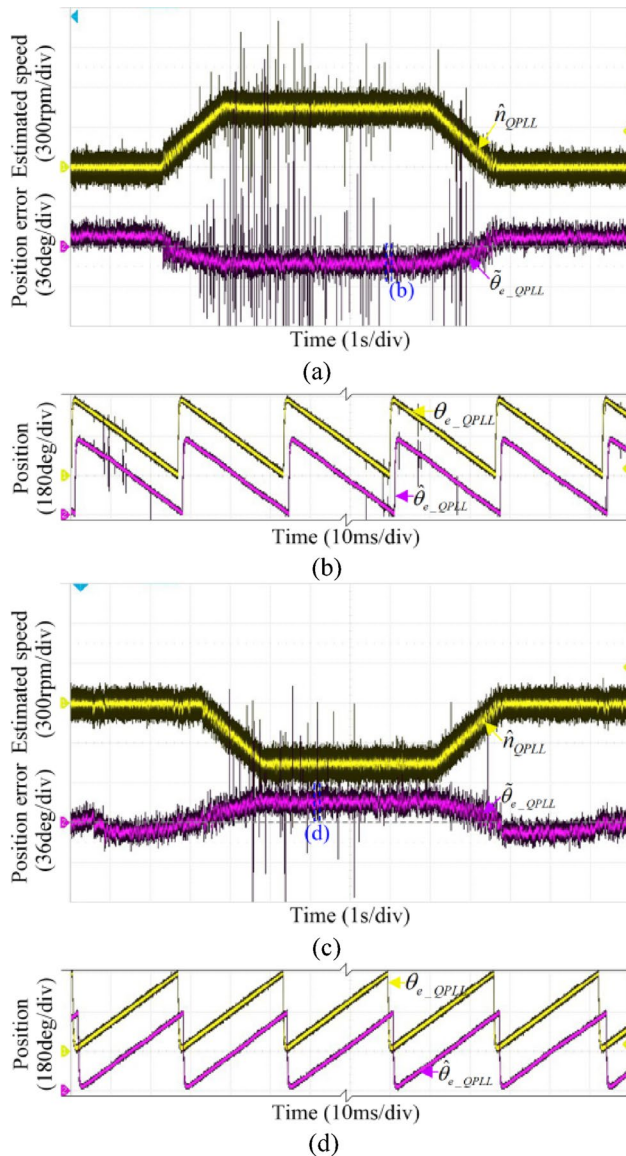


Fig. 14 Experimental result of QPLL system under rated load and switching speed between 0 and 450 rpm, as well as 0 and -450 rpm: **a-c** estimated speed and position error; **b-d** actual and estimated position

under no- or rated-load cases, the position error can be reduced to around 0, and only a certain transient error occurs near zero speed, with a maximum of 8.6° . In Fig. 16, the position error is 4.6° before and after suddenly unloading the rated load. The changed position of the rotor movement is 50.4° , which is equivalent to the system based on QPLL. The position waveforms with crossing zero speed and rated load is shown in Fig. 17e, which proves the excellent suppression capability of EPLL to position error.

In summary, by adopting HFPI-RS, the harsh spike harmonics in the current can be smoothed, which reduces the impact of noise on the environment. Table 3 lists the

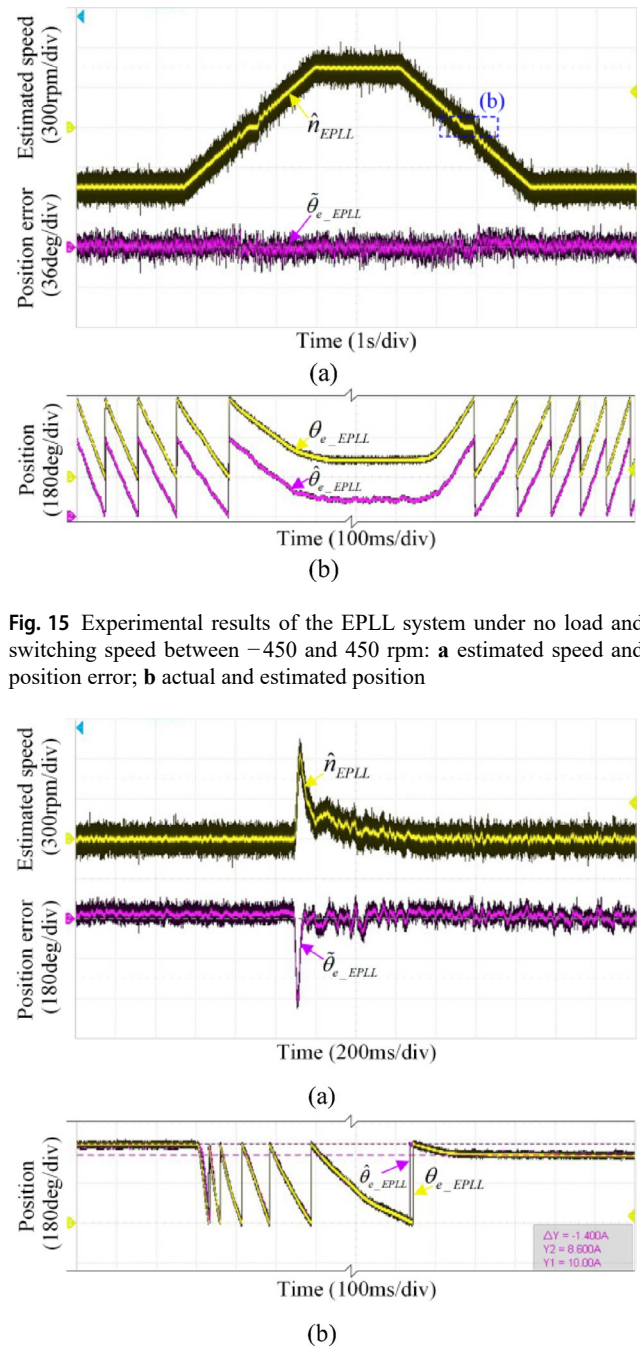


Fig. 15 Experimental results of the EPLL system under no load and switching speed between -450 and 450 rpm: **a** estimated speed and position error; **b** actual and estimated position

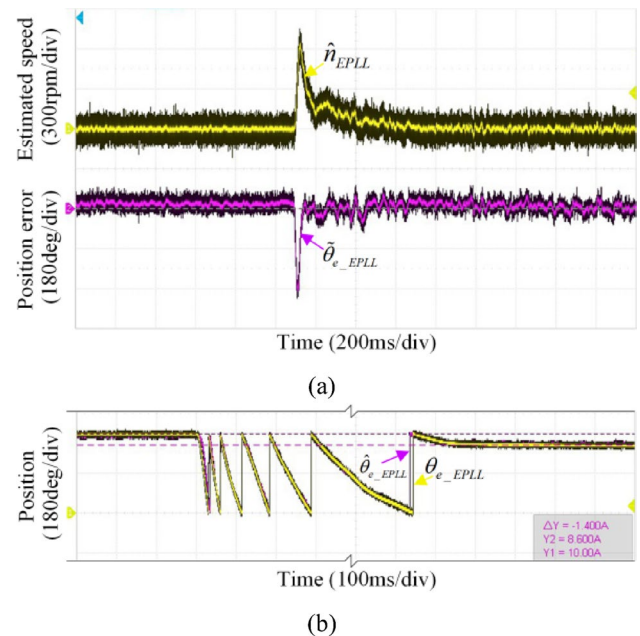


Fig. 16 Experimental result of the EPLL system with suddenly unloading rated load at zero speed: **a** estimated speed and position error; **b** actual and estimated position

maximum position errors under various operating conditions for an intuitive comparison of the performance differences between the QPLL- and EPLL-based sensorless systems. The proposed EPLL significantly improves the position error.

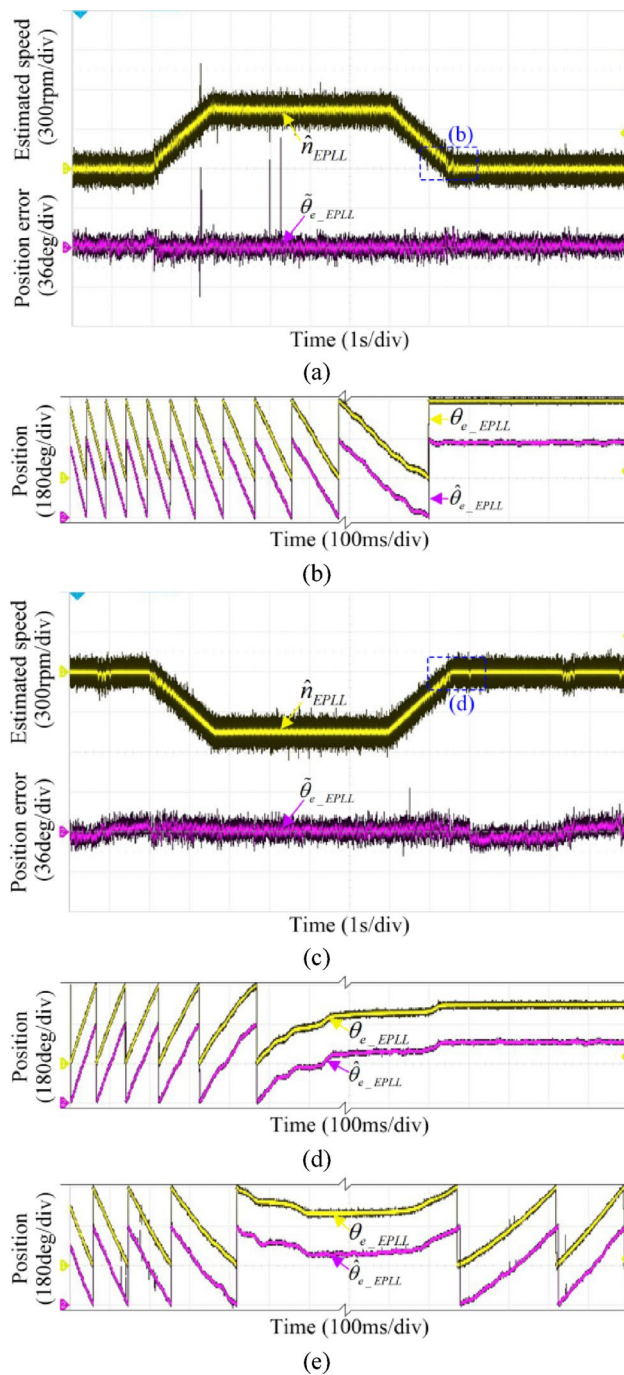


Fig. 17 Experimental result of the EPLL system under rated load and switching speed between 0 and 450 rpm, as well as 0 and -450 rpm: **a-c** estimated speed and position error; **b-d** actual and estimated position

Table 3 Comparison of maximum position errors based on QPLL and EPLL under various operating conditions

Sensorless system	No load	Rated load	Suddenly unloading the rated load at zero speed
QPLL-based	20°	17.4°	5.6°
EPLL-based	≈0°	≈0°	4.6°

5 Conclusion

For the sensorless system based on HF signal injection in the zero-low speed domain, harsh audible noise and large errors occur in the stage of given ramp speed. To address the two problems, this study proposes respective optimization methods. A pair of complementary random HF pulse injection is utilized to remove the harsh discrete term in the responsive current. In this way, the noise becomes stable and low-pitched. Moreover, an enhanced Type III PLL with simple structure and easy parameter tuning is proposed, which can weaken the transient error to zero. The two optimization measures greatly improve the dynamic performance and control accuracy of sensorless control systems.

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Declarations

Conflict of interest On behalf of all the authors, the corresponding author states no conflict of interest.

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